

Origins of Matrix/Vector and Matrix/Matrix Multiplication

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1 Generalizing $a x = b$

Why is matrix/vector and matrix/matrix multiplication defined the way it is? One motivation is to try to extend the (albeit trivial) solution of linear equations in 1-dimension to n dimensions. To do this, we go through the solution of the 1-dimensional case in careful detail.

The scalar problem is:

$$a x = b \tag{1}$$

Here is a very explicit solution keeping an eye towards generalization.

$$a x = b \tag{2}$$

$$a^{-1}(a x) = a^{-1}b \quad (\text{Multiply by Inverse}) \tag{3}$$

$$(a^{-1}a)x = a^{-1}b \quad (\text{Use Associativity of multiplication}) \tag{4}$$

$$1 x = a^{-1}b \quad (\text{What Inverse multiplication does}) \tag{5}$$

$$x = a^{-1}b \quad (\text{What Identity multiplication does}) \tag{6}$$

The multi-dimensional problem has a lot more variables and coefficients. To start, we need to be more systematic about the naming of these coefficients. With this in mind, the multi-dimensional linear problem can be written:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Here, a_{ij} is the coefficient in the i^{th} row and j^{th} column.

To make this look like the 1-dimensional case, we need to think of the b 's as a single unit. Our single unit will be the *vector* of the b 's.

The multi-dimensional case can be rewritten in vector terms as:

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (7)$$

We can replace the tedious list of sums on the left hand side of the above with standard mathematical summing notation:

$$\begin{bmatrix} \sum_{j=1}^n a_{1j}x_j \\ \sum_{j=1}^n a_{2j}x_j \\ \vdots \\ \sum_{j=1}^n a_{nj}x_j \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (8)$$

Two things need to be done: treat the x 's as a unit – as we did with the b 's – and separate the a coefficients from the x 's. This must be done formally while preserving the meaning of the original problem. This is done in the most natural of ways:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (9)$$

Taking the rectangular collection (matrix) of a 's to be represented by A ; the (vector) collection of x 's to be $\mathbf{x}$; and the collection of the b 's to be $\mathbf{b}$; we may write (9) as:¹

$$A\mathbf{x} = \mathbf{b} \quad (10)$$

Here, A represents the “matrix” of the a 's, with $A_{i,j} = a_{ij}$; meaning, that the i^{th} row and j^{th} column of A is a_{ij} . This now looks like the scalar problem, (1). Just as one writes $ax = b$, interpreting the proximity of a and x to mean multiplication, we want the proximity of A and $\mathbf{x}$ to mean matrix/vector multiplication – something we need to work out.

2 Solution of $A\mathbf{x} = \mathbf{b}$ determines Matrix Multiplication

We proceed to try solving (10) using the solution procedure used above for the scalar case. In the process, we need to:

- Vectorize the input variable x and the outputs b . – **Done**.
- Define an “ a ” matrix. – **Done**.

¹In general, we use uppercase italic letters for matrices; lowercase bold roman letters for vectors; and lowercase italic letters for scalars.

- Define matrix-vector multiplication.
- Define matrix-matrix multiplication.
- Define the identity matrix.
- Define the inverse matrix.

Here, in our new notation, is what the solution outline looks like for the multi-dimensional problem:

$$A \mathbf{x} = \mathbf{b} \quad (11)$$

$$A^{-1}(A \mathbf{x}) = A^{-1}\mathbf{b} \quad (\text{Matrix/vector multiply the vector equation by } A \text{ Inverse}) \quad (12)$$

$$(A^{-1}A)\mathbf{x} = A^{-1}\mathbf{b} \quad (\text{Use Associativity of matrix/matrix multiplication}) \quad (13)$$

$$I \mathbf{x} = A^{-1}\mathbf{b} \quad (\text{What Inverse matrix/matrix multiplication does}) \quad (14)$$

$$\mathbf{x} = A^{-1}\mathbf{b} \quad (\text{What Identity matrix/vector multiplication does}) \quad (15)$$

How do we make sense of this matrix/vector syntax on the left hand side of (11)? It should have the proper meaning; that is, (9) should have the same meaning as (8).

Examining (9) and the left hand side of (8), gives us our definition of matrix/vector multiplication:

$$[A \mathbf{x}]_i \equiv \sum_{j=1}^n A_{ij} x_j \quad (16)$$

That is, A acts on a vector, $\mathbf{x}$, to create a new vector, $A \mathbf{x}$, whose i^{th} entry is the i^{th} entry of the left hand side of (8).

Another way to write this is:

$$A \mathbf{x} = \sum_{j=1}^n x_j \mathbf{A}^j \quad (17)$$

Here, $\mathbf{A}^j$ is the j^{th} column vector of A .

This can be seen by writing (7) as:²

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{nn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (18)$$

From this we can see that matrix/vector multiplication of A on $\mathbf{x}$ is a “weighted” combination of the columns of A using the elements of $\mathbf{x}$ as the weights. Notice that if we pick an “ $\mathbf{x}$ ” with all of the weight concentrated on the j^{th} entry, then matrix/vector multiplication on this specific vector yields the j^{th} column of A , $\mathbf{A}^j$.

²Here, we assume the reader has a passing knowledge of how vectors are added and multiplied by scalars.

Specifically, applying A to the special vectors, $\mathbf{e}_j$ – which are 0 everywhere except at j where they are 1 – we see that $A\mathbf{e}_j = \mathbf{A}^j$. Consequently, matrix/vector multiplication on this set of vectors uniquely determines A – if there is another matrix that gives the same result for these vectors with respect to matrix/vector multiplication, its columns would have to match A .

We note the following for future reference.

$$[\mathbf{A}^j]_i = A_{i,j} \quad (19)$$

That is, the i^{th} entry of the column vector, $\mathbf{A}^j$, is the i^{th} row, j^{th} column of the matrix A .

Before moving on with our outline, we note one very important property of matrix/vector multiplication: It is *linear*. By that we mean the following:³

$$A(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha A\mathbf{x} + \beta A\mathbf{y} \quad (20)$$

One can show this from our definition of matrix/vector multiplication which we leave to the reader. What this means in practice is that for a given vector, $\mathbf{z} = \alpha\mathbf{x} + \beta\mathbf{y}$, we can compute $A\mathbf{z}$ by computing A on the “components” of $\mathbf{z}$ and then multiplying by scalars and adding the resulting vectors. This can be an easier way to compute the *action* of a matrix on a vector in some cases. The linearity result above can be extended to any finite linear combination:

$$A\left(\sum_{j=1}^m c_j \mathbf{x}_j\right) = \sum_{j=1}^m c_j A\mathbf{x}_j \quad (21)$$

Continuing with our solution outline, matrix/matrix multiplication is needed to go from equation (12) to (13). How must this be defined? Well, we need to make sense of:

$$(A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) \quad (22)$$

This involves the inverse of the matrix A , which we also need to define. We want associativity to hold *for all* matrix products, not just $A^{-1}A$. So we forgo what the inverse of a matrix might mean and focus on imposing the condition that matrix/matrix multiplication be associative for *all* input vectors. Specifically, the condition we impose is:

$$(BA)\mathbf{x} = B(A\mathbf{x}) \quad \forall \mathbf{x} \in \mathbf{R}^n \quad (23)$$

Notice that we are imposing what matrix/matrix multiplication is by saying how the new matrix formed, BA , acts on an *arbitrary* vector, $\mathbf{x}$. And we specify that action by the right hand side of (23), which involves only matrix/vector multiplication – something we have defined.

We want to emphasize that this is a strong condition to impose; meaning, that the requirement that this is true for *all* $\mathbf{x}$ gives us a lot to work with. For instance, if one had two $n \times n$ matrices, C and D , what

³You might ask: “How did someone think up such a strange property?” But it is not strange, it is what one might hope for, as all we are saying is that matrix/vector multiplication works the same way that ordinary *scalar* multiplication does – it *distributes* through addition and scalar multiplication.

could you conclude if someone told you that $C\mathbf{x} = D\mathbf{x}$ for *some* n vector, $\mathbf{x}$? Well, the answer is: not much. However, if I told you that $C\mathbf{x} = D\mathbf{x}$ for *all* $\mathbf{x}$, what could you say? I claim that tells us that $C \equiv D$; meaning that every entry in C is the same as the corresponding entry in D . We can see this by applying C and D to each of the $\mathbf{e}_j$, ($j \in [1, n]$) vectors. For each j , $C\mathbf{e}_j$ and $D\mathbf{e}_j$ pick off the j^{th} column of C and D respectively. So we see that each column of C and D must match; consequently, $C \equiv D$.

We apply this idea to (23) to find a formula for matrix/matrix multiplication. Since (23) must be true for *all* $\mathbf{x}$; in particular, it must be true for the vectors $\{\mathbf{e}_j\}_{j=1}^n$. This gives us the following equations:

$$(BA)\mathbf{e}_j = B(A\mathbf{e}_j) \quad j \in [1, n] \quad (24)$$

Or,

$$(BA)^j = B\mathbf{A}^j \quad j \in [1, n] \quad (25)$$

For any given, i , the i^{th} entry of the left and right hand side is (after using the definition of matrix/vector multiplication, (16), on the right hand side)

$$[(BA)^j]_i = \sum_{k=1}^n B_{i,k} [\mathbf{A}^j]_k \quad \forall i, j \in [1, n] \quad (26)$$

Using (19) on the left and right hand sides of the previous equation we have

$$(BA)_{i,j} = \sum_{k=1}^n B_{i,k} A_{k,j} \quad \forall i, j \in [1, n] \quad (27)$$

And we are done: we have shown what the new matrix BA is by giving every entry $(BA)_{i,j}$. So using the set of vectors, $\{\mathbf{e}_j\}_{j=1}^n$, we have completely determined what the matrix multiplication of BA is in order that matrix/matrix multiplication be associative when using these vectors. Does this mean that associativity works for *arbitrary* vectors? The answer is yes, by the linearity of matrix/vector multiplication. Any given $\mathbf{x}$ can be written as a *linear combination*⁴ of the vectors $\{\mathbf{e}_j\}_{j=1}^n$: $\mathbf{x} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n$. Matrix

⁴Again, we assume the reader is familiar with expanding a vector in terms of the standard basis vectors.

multiply the left and right hand sides by the matrix, $B A$. This gives

$$\begin{aligned}
(B A)\mathbf{x} &= (B A) \left(\sum_{j=1}^n a_j \mathbf{e}_j \right) && \text{Replace } \mathbf{x} \text{ with its components, } \mathbf{e}_j. \\
&= \sum_{j=1}^n a_j (B A) \mathbf{e}_j && \text{Linearity of matrix/vector multiplication by } (B A) \text{ (itself a matrix).} \\
&= \sum_{j=1}^n a_j B(A \mathbf{e}_j) && \text{Associativity of matrix/matrix multiplication on components, } \mathbf{e}_j. \\
&= B \left(\sum_{j=1}^n a_j A \mathbf{e}_j \right) && \text{Linearity of matrix/vector multiplication by } B. \\
&= B \left(A \left(\sum_{j=1}^n a_j \mathbf{e}_j \right) \right) && \text{Linearity of matrix/vector multiplication by } A. \\
&= B(A\mathbf{x}) && \text{Replace components with } \mathbf{x}.
\end{aligned} \tag{28}$$

In the above calculations, we applied our new matrix, $(B A)$ to an arbitrary $\mathbf{x}$, and used the *linearity* of matrix/vector multiplication and the *associativity* of matrix/matrix multiplication on the components, $\{\mathbf{e}_j\}_{j=1}^n$, to show that matrix/matrix multiplication, as we have defined it, is associative – *irrespective* of the input vector.

The next order of business is to identify an $n \times n$ matrix which serves as an identity (in terms of matrix/vector multiplication). Let us suppose that we have such a matrix and let's call it I . Then we must have $I \mathbf{x} = \mathbf{x} \quad \forall \mathbf{x} \in \mathbf{R}^n$. Then, in particular, $I \mathbf{e}_j = \mathbf{e}_j \quad \forall j \in [1, n]$. However, we know that $I \mathbf{e}_j = \mathbf{I}^j$. Therefore, I must have the property that $\mathbf{I}^j = \mathbf{e}_j$. Consequently, I must have the form:

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \tag{29}$$

We have shown that the only candidate for the identity matrix with respect to matrix/vector multiplication is the matrix I . That is, if there is an identity matrix, it must be the matrix of equation (29). Does it satisfy the property of being an identity matrix (again, in the matrix/vector multiplication world)? That is, do we have $I \mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbf{R}^n$? As before, we can express $\mathbf{x}$ as: $\mathbf{x} = \sum_{j=1}^n c_j \mathbf{e}_j$. By the *linearity* of

matrix/vector multiplication, we have

$$\begin{aligned}
I\mathbf{x} &= I \left(\sum_{j=1}^n c_j \mathbf{e}_j \right) && \text{Replace } \mathbf{x} \text{ with components, } \mathbf{e}_j. \\
&= \sum_{j=1}^n c_j I\mathbf{e}_j && \text{Linearity of matrix/vector multiplication.} \\
&= \sum_{j=1}^n c_j \mathbf{e}_j && \text{Identity of } I \text{ on components, } \mathbf{e}_j. \\
&= \mathbf{x} && \text{Replace components with } \mathbf{x}.
\end{aligned} \tag{30}$$

One can show that this identity matrix, I , is also the identity operator for matrix/matrix multiplication: applying $(IA)\mathbf{e}_j = I(A\mathbf{e}_j) = A\mathbf{e}_j$ to each j shows $(IA)^j = A^j$ for all j , hence $IA = A$; the same argument on the columns of A gives $AI = A$.

The only remaining piece is to determine when a matrix inverse exists and how to compute it. We do not attempt to do this in this paper. In the next section we continue with a qualitative comparison of the solution to the scalar problem, $ax = b$, and its vectorized cousin.

3 Qualitative Features of Solutions

We can view the multiplication of two numbers, a and x , as just that. Or, we can think of a being fixed and letting x “run through” all numbers. Here we see two cases: if $a \neq 0$, then letting x run through all of the numbers in $\mathbf{R}$ will produce all of the numbers in $\mathbf{R}$. We could think of a as an “operator” and call the set of all possible outputs, the range of a and denote it: $\mathcal{R}(a)$. There is another case: a could be zero. In this case its range is the set $\{0\}$. In the first case, with non-zero a it is clear that we can find an x to “hit” a given value b . That is, we can solve $ax = b$.

One can define the same concept for a matrix, A . In this case, we produce a “space” of all possible outputs of A as it acts on every input vector $\mathbf{x} \in \mathbf{R}^n$. This amounts to, using (17), the set of all *linear combinations* of the columns of A . Using this language of ranges, here is what we can say about the scalar problem: $ax = b$.

Unique Solution: If a^{-1} exists ($a \neq 0$), then there is a **unique** solution.

No Solution: If a^{-1} does not exist (i.e., $a = 0$) *AND* b is **not** in the range of a (that is: $b \neq 0$), then there is **no** solution.

Infinite Solutions: If a^{-1} does not exist (i.e., $a = 0$) *BUT* b is in the range of a (that is: $b = 0$), then there are an **infinite** number of solutions.

Here is the analog of this solution categorization for the multi-dimensional case: $A\mathbf{x} = \mathbf{b}$.

Unique Solution: If A^{-1} exists, then there is a **unique** solution.

No Solution: If A^{-1} does not exist *AND* $\mathbf{b}$ is not in the range of A (that is $\mathbf{b} \notin \mathcal{R}(A)$), then there is **no** solution.

Infinite Solutions: If A^{-1} does not exist *BUT* $\mathbf{b}$ is in the range of A (that is: $\mathbf{b} \in \mathcal{R}(A)$), then there are an **infinite** number of solutions.